Spectrometer (Prism and Diffraction Grating)

1 Aim

1. To find the prism angle and the angle of minimum deviation of the prism.
2. To determine the refractive indices of a glass prism as a function of wavelengths of mercury light, by refraction of light through prism at minimum deviation.
3. To plot the dispersion curve for the glass prism and calculate the dispersive power of the prism.
4. To obtain the coefficients in Cauchys equation from the graph of n versus $1/\lambda^2$.
5. To use a diffraction grating to determine wavelength of sodium light.

2 Introduction

An optical instrument used to measure the wavelengths or frequencies of light emitted by various light sources is known as a spectrometer. All spectrometers depend on an optical element, usually a prism or a grating, to separate the light into its individual colors or wavelengths. Once separated, the different wavelengths can be measured. The refractive index of the prism material depends on the wavelength of light and can be written in terms of the Cauchy equation as

$$n = A + \frac{B}{\lambda} + \frac{C}{\lambda^4} + ...$$

where the constant A is the coefficient of refraction and B is the coefficient of dispersion.

The prism spectrometer consists of a collimator, a prism table and a telescope. Light enters through a slit at one end of the collimator and emerges parallel to the axis of the collimator at the other end. The collimated light then strikes a prism that has been placed on the spectrometer table. The prism deviates and disperses the collimated light according to Snells law and the different wavelengths of light present in the light source.

The sets of parallel rays (one set for each wavelength) then enter the telescope and form separate distinct images of the collimator slit when viewed through the telescope. The angular position of the telescope is determined with a circular scale, which is graduated in fractions of a degree. A distinct relationship exists between the position of the telescope and a given wavelength of light passing through the prism. This relationship permits the prism spectrometer to be used to measure the wavelength of any given spectral light component.

When a ray of monochromatic light passes through a prism, it is refracted twice, once as it enters and again as it leaves the prism. The angle A between the two prism surfaces or planes where refraction takes place is called the prism angle. The intersection of the two refracting planes is called the refracting edge. The base of the prism is that side of the prism opposite the prism angle. The incident and refracted rays are in different directions and the angle D between the incident and emergent rays is defined as the angle of deviation. If a beam of collimated light containing several wavelengths is incident on the prism face, different wavelengths will be refracted by different amounts and will have different angles of deviation. The particular angle of deviation depends on the prism angle, the index of refraction of the prism at that wavelength and the angle of incidence. The angle of deviation D is a minimum if the angle of emergence is equal to the angle of incidence. When the prism spectrometer is set at minimum deviation for a given wavelength, the index of refraction n (at that wavelength) is given by

$$m = \frac{\sin(A + D_{min})/2}{\sin A/2}$$

where A is the prism angle, D_{min} is the measured angle of minimum deviation for the particular wavelength. By determining the angle of minimum deviation for each wavelength, one can calculate the appropriate refractive index. The graph of index of refraction versus wavelength is called a dispersion curve.

From the dispersion curves for different optical materials it can be seen that the index of refraction varies with the wavelength. This rate of change of index of refraction with wavelength is called dispersion and can be written as $\Delta n/\Delta\lambda$. The relative dispersion or dispersive power, DP, of a prism in the visible region is defined as

$$DP = \frac{n_v - n_y}{n_{avg} - 1},$$

where n_v and n_y are refractive indices for voplet and yellow light respectively and n_{avg} is their average. The resolving power of a prism is

$$RP = B\frac{\Delta n}{\Delta\lambda},$$

where B is the length of the base of the prism.

Diffraction grating is an optical component with a periodic structure (a glass plate with rulings which act as slits) that splits and diffracts light into several beams travelling

Figure 1: Spectrometer with prism and a grating.

in different directions. The directions of these beams depend on the spacing of the grating and the wavelength of the light.

When light is normally incident on the grating, the diffracted light has maxima at angles θ_n given by

$$d \sin \theta_n = n\lambda$$

where d is the distance from the center of one slit to the center of the adjacent slit, n denotes number of maxima.

3 Apparatus in the lab and procedure

3.1 Adjustment of prism table and Schusters method

1. Level the prism table using a spirit level and leveling screws.

2. Before placing the prism on the table, aim the telescope at a distant object through a window and bring the image of the distant object into clear focus in the plane of the cross hairs (focal plane of the telescope).

3. Next, adjust the collimator for parallel light. Bring the telescope into a straight line with the collimator.

4. Open the adjustable slit. Place a light source in front of the slit and look through the telescope. Slide the tube that holds the slit in and out of the barrel of the collimator and, if necessary, move the telescope sideways a little until a sharp image of the slit is seen in the center of the focal plane of the telescope.

5. Using the spirit level and adjusting screws, the prism table should be adjusted in horizontal plane. The prism table is adjusted to be at the same height of the collimator and telescope.

6. Now the prism is placed on the prism table in such a position that one of the refracting surfaces faces the collimator. The light emerging from the other refracting surface of the prism is viewed through the telescope.

7. The prism table and the telescope both are then rotated slowly and simultaneously so that the spectral lines always remain at the cross-wire. A state is reached when on rotating the prism table further, the direction of rotation of the spectral lines is reversed. This position is the position of minimum deviation for that particular wavelength (where the angle of incidence will be equal to the angle of emergence). Determine this position of minimum deviation very carefully by using the fine adjustment screws on the telescope. Perform this procedure several times and take the average of your values for the minimum deviation position.

8. Now keeping the telescope fixed, the prism table is rotated through a small angle. The spectral lines are seen to be blurred. The telescope is adjusted so that spectral lines, become distinct and clear.

9. Now the prism table is rotated through a small angle in the reverse direction. Again the spectral lines are seen blurred. The collimator is adjusted so that spectral lines become distinct and clear.

10. The methods described in the previous steps are repeated again and again till the spectral lines become distinct and clear throughout the entire rotation of the telescope. In this position the spectrometer will be set for parallel rays coming out of the collimator, focusing parallel rays by telescope at its cross-wire and the prism in the position of angle of minimum deviation.

3.2 Measurement of the prism angle

1. Place the prism on the table of the spectrometer so that its refracting edge is vertical and faces the collimator.

2. Move the telescope until the collimator slit image, reflected off one of the prism faces, is visible in the telescope. Carefully position the telescope until the slit image is in the center of the focal plane of the telescope and the cross hair is aligned with one edge of the slit image. Read and record the angular position of the telescope on the graduated scale, making use of the vernier to obtain maximum accuracy.

3. Swing the telescope around to receive the light reflected off the other face of the prism. (When rotating the telescope, push on the supporting arm rather than on the telescope itself.) Align the cross hair with the same edge of the slit used in position in the point above. Read and record the angular position of the telescope. The difference in readings of the two angular positions is equal to twice the prism angle.

3.3 Measurement of the angle of minimum deviation

1. Turn on a mercury lamp source. After the mercury lamp has brightened to its highest intensity (several minutes), set the lamp near the slit end of the collimator. With clamps loosened, swing the telescope around in line with the collimator. With the prism removed, center the slit image in the focal plane of the telescope. Make the slit fairly narrow, but not so narrow that a single-slit diffraction pattern occurs.

2. Read and record the angular position of the telescope.

3. Place the prism on the spectrometer table so that one face makes an angle of about 45^p with the light coming from the collimator. At the same time, ensure that the collimator light fills the face of the prism. Look through the telescope while moving it around slowly, until the colored images of the collimator slit are seen. Align the cross hair of the telescope on one of the spectral lines.

4. To find the position of minimum deviation, the prism table is rotated slowly in the direction that causes the image of the bright line to move toward the first position of the telescope, that is, toward the undeviated direction. Follow this image with the telescope until a position is found where the image just reverses its motion. Repeat this several times until the turn-around point is clearly determined. This is the direction of minimum deviation for the line.

5. Carefully align the cross hair on one of the spectral lines and read the angular position of the telescope. The difference between this reading and the reading for the undeviated direction is equal to the angle D_{min} .

6. Compute the value of the refractive index n.

3.4 Diffraction Grating

(a) Set the grating normal to the incident light.

(b) Measure the positions of the maxima on both sides (left we well as right) for different wavelengths of light.

(c) Compute the wavelength of light using the grating formula.

4 Precautions

1. The telecope and collimator settings should not be changed once they have been focussed appropriately.

2. Rotate the spectrometer table only in one direction to avoid back lash error.

3. Make sure that you wait for a few minutes before taking readings so that your eyes are adjusted to the dark.